Vidyavardhini's College of Engineering & Technology
(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & DTE)
NBA & NAAC Accredited
Department of Computer Science Engineering (Data Science)

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2026-27

Course: Agile Software Development and DevOps Lab

Course Code: 2015119

Year: TE SEM: V

Experiment No. 1
AIM:- To study and implement the Agile Software Project Lifecycle using Jira and integrate it with a DevOps CI/CD pipeline.
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Roll Number: 50
Date of Performance: 21/9/26
Date of Submission: 21/9/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by

Name of Faculty : Asst. Prof. Shilpa Bane

Signature :

Date :

To study and implement the Agile Software Project Lifecycle using Jira and integrate it with a DevOps CI/CD pipeline.

Theory:

Agile Software Development:

Agile is an iterative and incremental approach to software development. In Agile, the software project is divided into small units of work called **Sprints**. Each Sprint delivers a working part of the software.

The main principles of Agile are:

1. Iterative and incremental development.
2. Continuous customer feedback.
3. Team collaboration.
4. Adaptation to changing requirements.
5. Continuous testing and improvement.

Agile Project Lifecycle:

The basic Agile project lifecycle consists of:

Requirement → Product Backlog → Sprint Planning → Development → Testing → Review → Retrospective → Release

Jira:

Jira is a project management and issue tracking tool used for managing Agile software projects.

Important Jira components are:

1. **Project:** Container for all project activities.
2. **Epic:** Large feature or module of the project.
3. **User Story:** Requirement written from the user's point of view.
4. **Task:** Specific work that has to be performed.
5. **Bug:** Defect identified in the software.
6. **Product Backlog:** List of all pending requirements and tasks.
7. **Sprint:** A fixed development period.
8. **Scrum Board:** Board used to track the progress of issues.

DevOps:

DevOps is a software development practice that combines **Development and Operations**. It uses automation to improve the speed and reliability of software development and delivery.

CI/CD Pipeline:

CI/CD stands for **Continuous Integration and Continuous Delivery/Deployment**.

A CI/CD pipeline automatically performs activities such as:

Code → Build → Test → Package → Deploy

When a developer pushes code to the repository, the pipeline can automatically build and test the application.

Jira and DevOps Integration:

Jira can be used with Git and CI/CD tools to track the complete software development process.

For example, a Jira issue may have the key:

JIRA-101

The developer can include this key in the Git commit message:

JIRA-101 Implement login functionality

This provides traceability between the Jira requirement and the corresponding source-code change.

Steps:

1. Create Jira Project

1. Open Jira and log in.
2. Create a new **Software Project**.
3. Select the **Scrum** template.
4. Enter the project name, for example:

Online Book Store

5. Create the project.
6. Open the Scrum Board.

2. Create Product Backlog

Create an Epic:

Epic: Online Book Store

Create the following User Stories:

Issue	Type	User Story	Points
OBS-01	Story	As a user, I want to register an account	3
OBS-02	Story	As a user, I want to login	3
OBS-03	Story	As a user, I want to view books	5
OBS-04	Story	As a user, I want to search books	5
OBS-05	Task	Create database for books	3
OBS-06	Bug	Fix login validation error	2

3. Create Sprint

1. Open the **Backlog** section.
2. Create **Sprint 1**.
3. Select the required User Stories and Tasks.
4. Set the Sprint Goal.

Example:

Sprint Goal:
Implement user login and book browsing functionality.

5. Start the Sprint.

4. Manage Issues Using Scrum Board

Move issues through the following workflow:

To Do → In Progress → Testing → Done

For example:

OBS-02
↓
To Do
↓
In Progress
↓
Testing
↓
Done

5. Create Git Repository

1. Create a Git repository for the project.
2. Create or add the application source code.
3. Initialize

Git. git init

4. Add the project files.

git add .

5. Commit the code.

git commit -m "OBS-02 Implement login functionality"

6. Push the code to the remote

repository. git push origin main

6. Integrate Jira with Development

While developing a Jira User Story, use its issue key in the Git commit message.

Example:

git commit -m "OBS-03 Implement book browsing"

This helps identify which code changes belong to which Jira User Story.

7. Configure DevOps Pipeline

Create a CI/CD workflow for the project.
The pipeline should contain the following stages:
Checkout
↓
Build
↓
Test
↓
Package
↓
Deploy

For a Java Maven project, create:

.github/workflows/ci.yml

Use the following workflow:

```
name: Java CI

on:
  push:
    branches:
      - main

jobs:
  build:
    runs-on:

    ubuntu-latest steps:
```

```
- name: Checkout Code
  uses: actions/checkout@v4

- name: Setup Java
  uses: actions/setup-java@v4
  with:
    distribution: temurin
    java-version: '17'

- name: Build and Test
  run: mvn clean test

- name: Package
  run: mvn package -DskipTests
```

8. Execute CI Pipeline

1. Push the workflow file to the repository.
2. Make a change to the source code.
3. Commit the change using the Jira issue key.

```
git add .
git commit -m "OBS-04 Implement book search"
git push origin main
```

4. Open the repository's **Actions/Pipeline** section.
5. Observe the execution of the pipeline.
6. Verify the following stages:

✓ Checkout Code
✓ Build
✓ Test
✓ Package

7. If all stages are successful, the build is considered ready for deployment.

9. Sprint Review

At the end of the Sprint:

1. Open the Jira Scrum Board.
2. Verify completed User Stories.
3. Demonstrate the implemented application.
4. Show the successful CI/CD pipeline.
5. Review the Sprint progress.

10. Sprint Retrospective

Discuss:

1. What went well?
2. What problems were encountered?
3. What can be improved in the next Sprint?

Output:

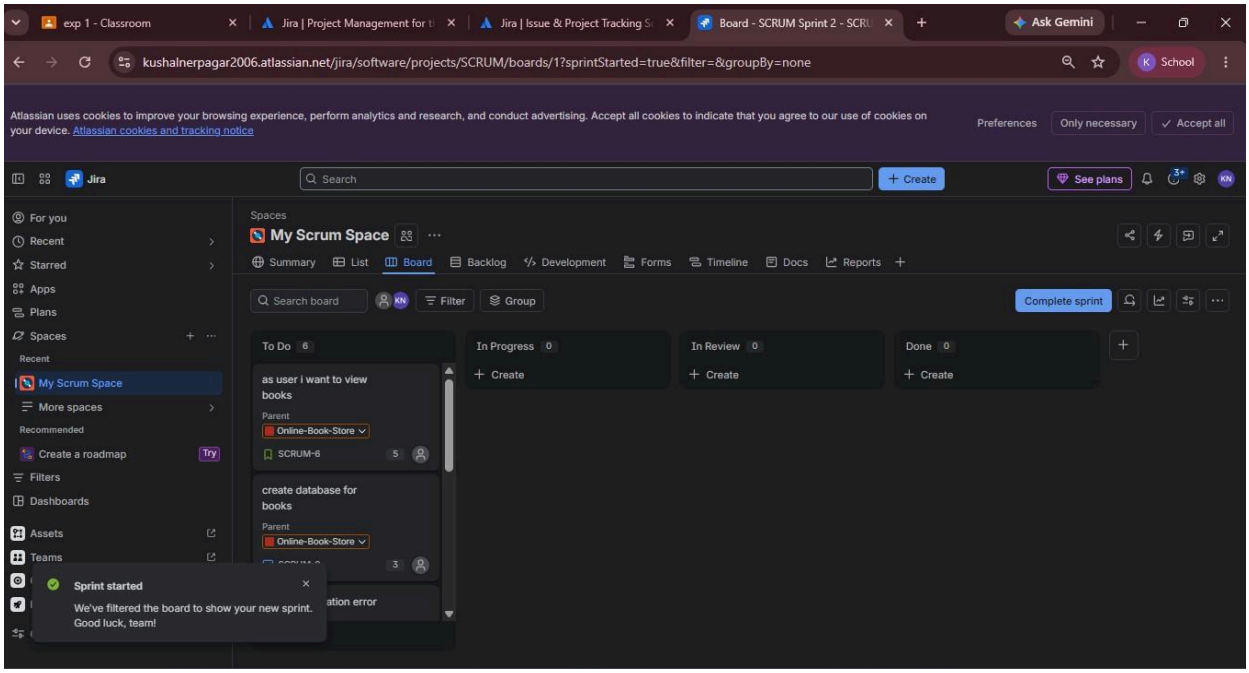
The following output should be obtained:

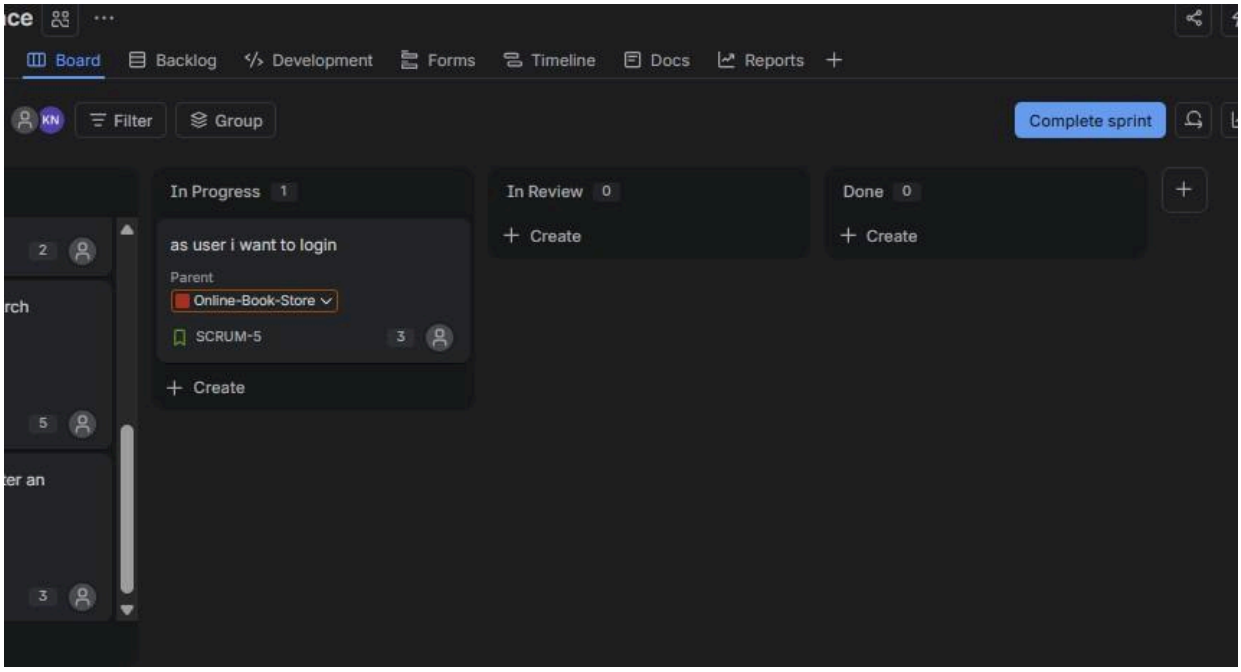
1. Jira Scrum project created successfully.
2. Product Backlog created with Epics, User Stories, Tasks, and Bugs.
3. Sprint created and started successfully.
4. Issues moved through:

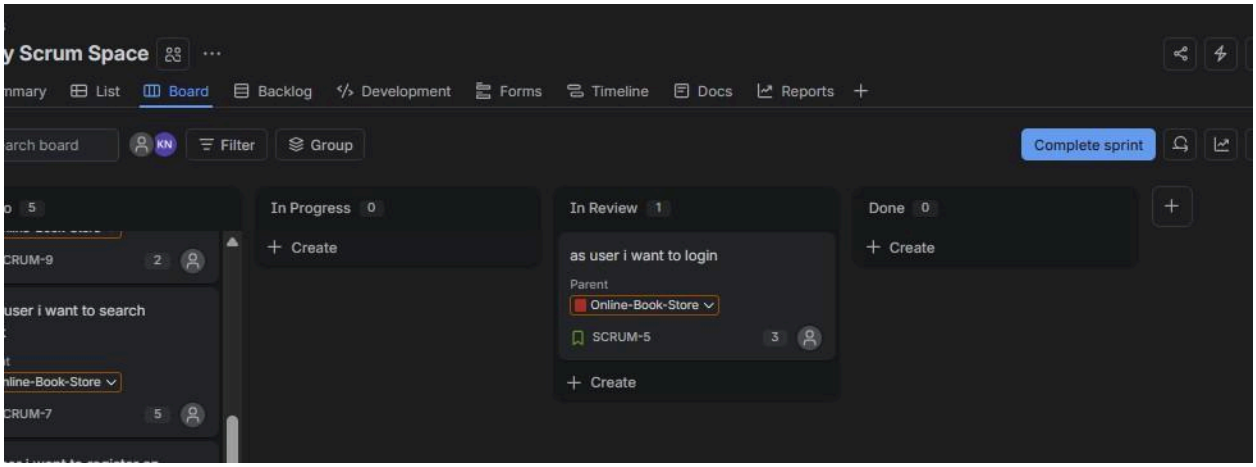
To Do → In Progress → Testing → Done

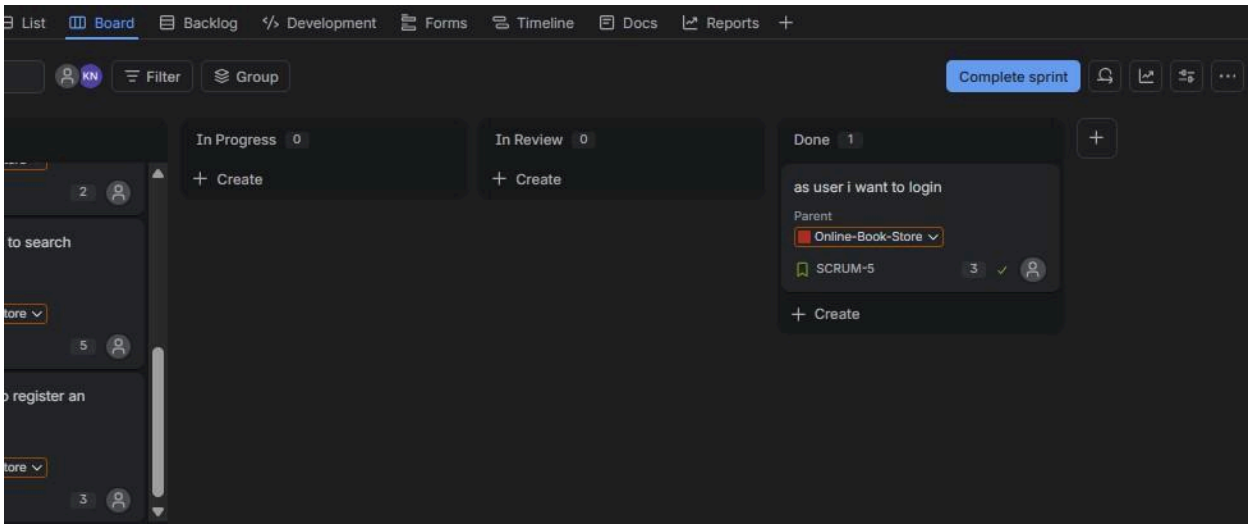
5. Git repository created successfully.
6. Jira issue keys linked with Git commit messages.
7. CI/CD pipeline configured successfully.
8. Source code built and tested automatically.
9. Successful pipeline execution displayed.
10. Agile and DevOps lifecycle successfully demonstrated.

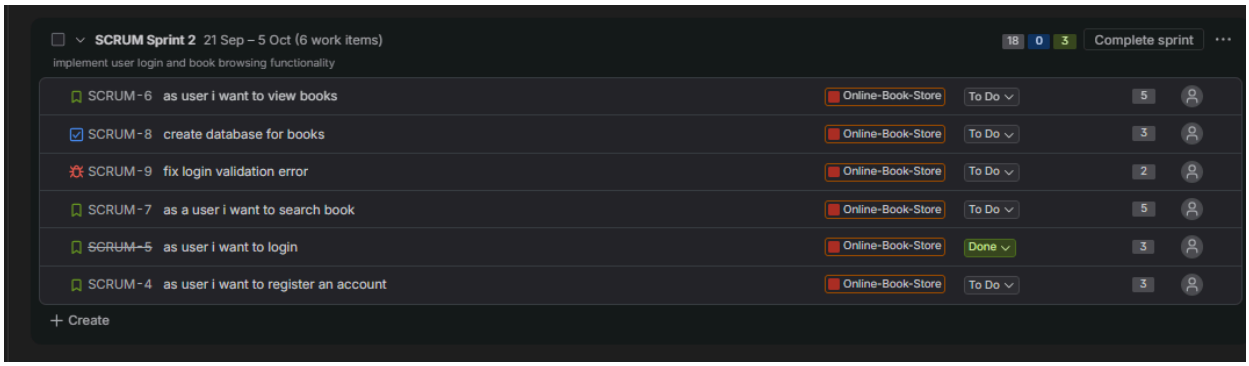
Result :-

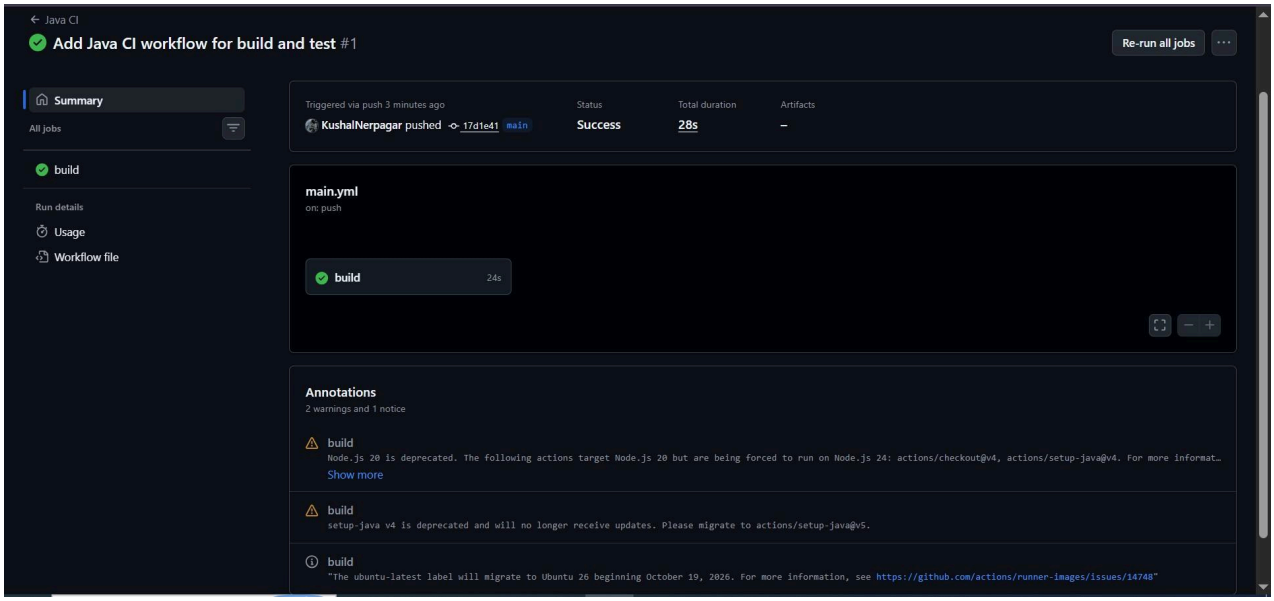

exp 1 - Classroom
Jira | Project Management for t
Jira | Issue & Project Tracking S
Board - SCRUM Sprint 2 - SCRU
Ask Gemini
kushalnerpagar2006.atlassian.net/jira/software/projects/SCRUM/boards/1?sprintStarted=true&filter=&groupBy=none
School
Atlassian uses cookies to improve your browsing experience, perform analytics and research, and conduct advertising. Accept all cookies to indicate that you agree to our use of cookies on your device. Atlassian cookies and tracking notice
Preferences
Only necessary
Accept all
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
My Scrum Space
More spaces
Recommended
Create a roadmap
Try
Filters
Dashboards
Assets
Teams
Spaces
My Scrum Space
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Search board
Filter
Group
Complete sprint
To Do 6
In Progress 0
In Review 0
Done 0
Create
as user i want to view books
Parent
Online-Book-Store
SCRUM-6
5
create database for books
Parent
Online-Book-Store
3
Sprint started
We've filtered the board to show your new sprint. Good luck, team!

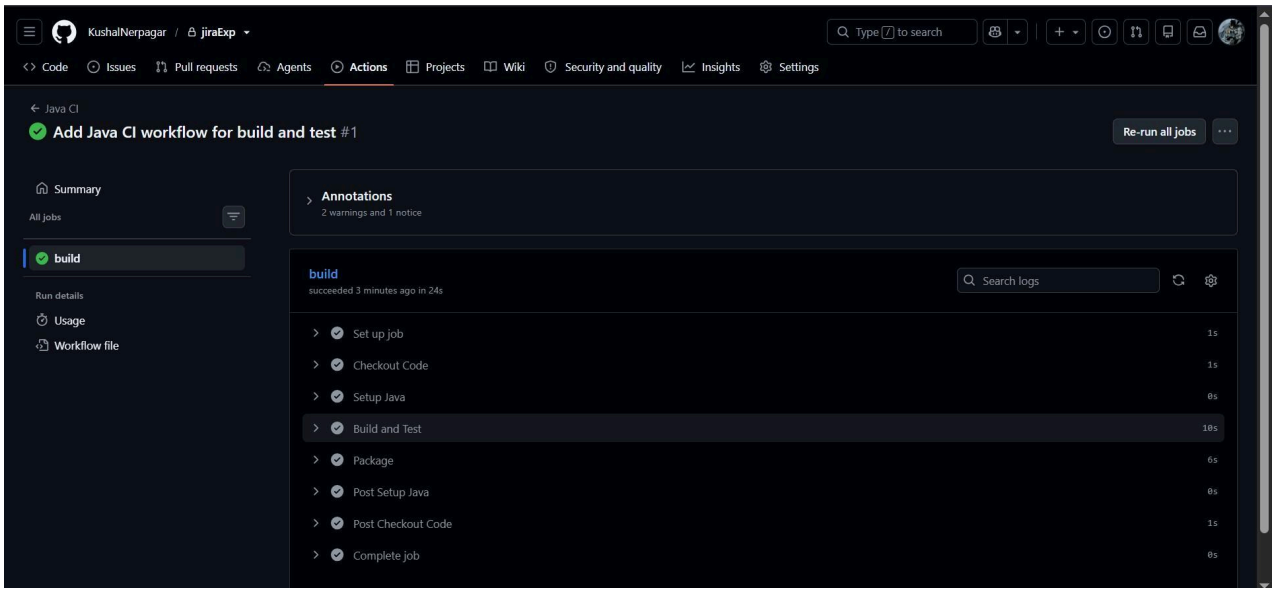

Board
Backlog
Development
Forms
Timeline
Docs
Reports
Filter
Group
Complete sprint
In Progress 1
In Review 0
Done 0
as user i want to login
Parent
Online-Book-Store
SCRUM-5
3
Create
Create
Create


y Scrum Space
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Filter
Group
Complete sprint
In Progress 0
Create
In Review 1
as user i want to login
Parent
Online-Book-Store
SCRUM-5
3
Create
Done 0
Create
user i want to search
SCRUM-7
5


List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Filter
Group
Complete sprint
In Progress 0
Create
In Review 0
Create
Done 1
as user i want to login
Parent
Online-Book-Store
SCRUM-5
3
Create
to search
register an


SCRUM Sprint 2 21 Sep – 5 Oct (6 work items)
implement user login and book browsing functionality
18 0 3
Complete sprint
SCRUM-6 as user i want to view books Online-Book-Store To Do 5
SCRUM-8 create database for books Online-Book-Store To Do 3
SCRUM-9 fix login validation error Online-Book-Store To Do 2
SCRUM-7 as a user i want to search book Online-Book-Store To Do 5
SCRUM-5 as user i want to login Online-Book-Store Done 3
SCRUM-4 as user i want to register an account Online-Book-Store To Do 3
Create





Conclusion:

Thus, the agile software project lifecycle was successfully implemented using jira, and the project was integrated with git and github actions. The jira backlog and sprint were managed successfully, and the CI/CD pipeline automatically performed the build, testing, and packaging stages.